

Experient 1 Private Cloud using Openstack

Aim:

To design, install, and configure a Private Cloud environment using a single-node OpenStack (MicroStack) implementation, and to understand the basic cloud services such as compute, networking, and storage by deploying and accessing virtual machine instances on the private cloud.

Openstack:

What is openstack?

1. OpenStack is a free, open standard cloud computing platform.
2. It is mostly deployed as infrastructure-as-a-service(IaaS)
3. OpenStack allows to deploy virtual machines (VMs) on demand.

It controls large pools of

1. compute 2) storage 3) networking resources of a datacentre

-managed by admins via dashboard -provides resources to users via web interface.

Private/Hybrid cloud->OpenStack. Public cloud ->AWS,Azure... **what**

is microstack?

MicroStack runs OpenStack locally with minimal setup(single-node).

Ideal for learning, testing, or small development projects.

It's easy to deploy and test OpenStack on a single machine. (like a laptop or small server). **openstack services:**

1. Nova (Compute)
2. Neutron (Networking)
3. Keystone (Identity)
4. Glance (Images)
5. Horizon (Dashboard)

Function	OpenStack	AWS	Description
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Compute (VMs)	Nova	EC2	Run virtual machines
Image Management	Glance	AMI	Store and manage VM images
Identity & Access	Keystone	IAM	User authentication, authorization, roles
Networking	Neutron	VPC, SecurityGroups	Manage networks, routers, IPs, firewalls, load balancing
Block Storage	Cinder	EBS	Persistent volumes for VMs
Object Storage	Swift	S3	Store and retrieve large amounts of unstructured data
Dashboard (UI)	Horizon	AWS Console	Web interface for managing cloud resources

[OPENSTACK\(microstack\) installation steps in Ubuntu 22.04 LTS\(jammy\)](#)

sudo snap install microstack --beta snap list microstack sudo

apt install python-is-python3 sudo microstack init --auto --

control type🗎**open browser and type 10.20.20.1 dashboard**

appears

username=admin

//for password

sudo snap get microstack config.credentials.keystone-password

1. MicroStack Installation

```
student@ubuntu:~$ sudo snap install microstack --beta
microstack 2024.1.1 from MicroStack✓ installed

student@ubuntu:~$ snap list microstack
Name      Version  Rev   Tracking   Publisher   Notes
microstack 2024.1.1 4066  latest/beta canonical✓  -
```

2. Python Support Installation

```
student@ubuntu:~$ sudo apt install python-is-python3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python-is-python3 is already the newest version (3.10.6-1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

3. Initialize MicroStack

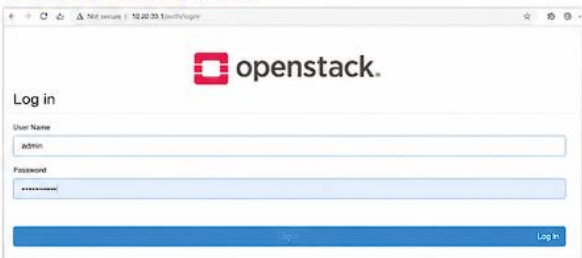
```
student@ubuntu:~$ sudo microstack init --auto --control
[INFO] Waiting for snap "microstack" to fully install      done
[INFO] Configuring...                                       done
[INFO] Starting MicroStack services ...                     done
[INFO] MicroStack is ready.                                done

student@ubuntu:~$
```

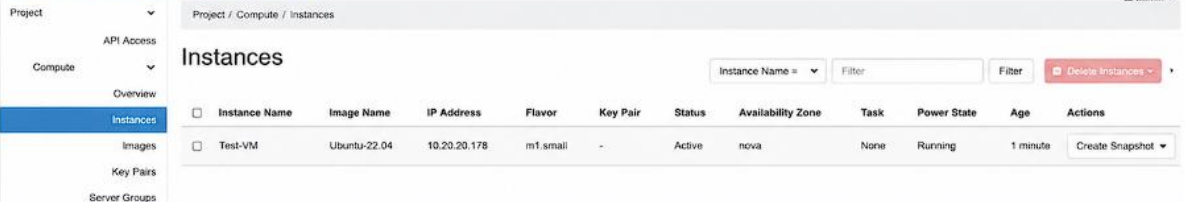
5. Get Keystone Password

```
student@ubuntu:~$ sudo snap get microstack \
config.credentials.keystone-password
config.credentials.keystone-password: 4Qp7x9Z7n8eK
student@ubuntu:~$
```

4. Access Dashboard (Horizon)



6. Instance (VM) Created Successfully



← → ↺

🔒 10.20.20.1/project/

 **openstack.**

admin ▾

Project ▾

API Access

Compute ▾

Overview

Instances

Images

Key Pairs

Server Groups

Volumes >

Network >

Admin >

Identity >

Project / Compute / Overview

Overview

Limit Summary

Compute

Instances

Used 0 of 10

VCPUs

Used 0 of 20

RAM

Used 0Bytes of 50GB

Volume

Volumes

Volume Snapshots

Volume Storage

Result

A single-node Private Cloud environment was successfully set up using OpenStack (MicroStack). The OpenStack services were properly installed and configured, and the private cloud was able to launch, manage, and access virtual machine instances with functional compute, networking, and storage services, thereby meeting the objectives of the experiment.